

Conchordal

Emergent Harmony via Direct Cognitive Coupling
in a Psychoacoustic Landscape

Koichi Takahashi

Institute for Advanced Biosciences and Graduate School of Media and Governance, Keio University, Japan
Conchordal.org · ktakahashi@conchordal.org

Sound as an ALife substrate

Spatial

cellular automata,
morphogenesis

Chemical

autocatalysis,
artificial chemistry

Symbolic

genetic programming,
rule systems

Acoustic

underexplored as a
primary medium

THE SCIENTIFIC QUESTION

Can a landscape derived from **psychoacoustic observables** function as a non-traditional ALife substrate — supporting **self-organization, selection, heredity, and synchronization**?

Sound has rich physical structure and is deeply entangled with biological perception — yet it remains underexplored as a computational medium for life-like dynamics.

Why must it be this sound?

Pythagoras

6th c. BC

Harmony as whole-number ratios.

Rameau

1722

Harmony from the overtone series.

Schoenberg

1908

All twelve pitches made equal.

Grisey · Murail

1970s

Composing from the measured spectrum.

Takemitsu

1990s

“A sea of tonality” — the pull of sound itself.

*Since Pythagoras, harmony has been sought in number. **None of it answered the question.***



Direct Cognitive Coupling (DCC)

For perceptually grounded generative systems.

(i) Perceptual axes

Axes computed from perceptual observables: cochlear roughness, periodicity-based harmonicity.

(ii) Direct read-out

Agents read that landscape directly. No symbolic rules, no scales, no trained parameters.

WHAT THE PRINCIPLE CLAIMS

Ecological–Aesthetic Duality

One landscape governs survival and turnover. The same field is the proxy for musical coherence.

A claim about model structure. Human judgement not tested here.

WHAT THE ASSAYS TEST

The scientific question

Can a landscape derived from **psychoacoustic observables** function as a non-traditional ALife substrate — supporting **self-organization, selection, heredity, and synchronization**?

The same question, now under test in four assays.

The diagram illustrates the structure of the cochlea. On the left, a spiral view shows the cochlear duct and the location of the stapes, oval window, and round window. An arrow points to a longitudinal section on the right. This section shows the stapes moving at the oval window, the round window, the base and apex of the cochlea, the basilar membrane, and the two scalae: the Scala vestibuli (top) and Scala tympani (bottom). A frequency spectrum is shown at the top, ranging from 20 kHz on the left to 20 Hz on the right, with a label 'frequency sensitivity'. A wave on the basilar membrane is shown, with a label 'correspond to loudness' pointing to its amplitude. The helicotrema is indicated at the apex of the cochlea.

Tonotopy

The membrane is stiff and narrow at the stapes end, wide and loose at the apex, so each place responds best to one frequency.

Why partials interact

Two partials close in frequency excite overlapping places. Inside one critical band the ear cannot separate them, and the residue is heard as roughness.

Why this matters here

The landscape's axes are not musical conventions. They are consequences of where sound lands on this membrane.

THE LANDSCAPE AXIS · R_{01} , normalised to $[0, 1]$

WHERE IT COMES FROM · early peripheral processing, rooted in cochlear mechanics

Two partials sum to a modulated envelope

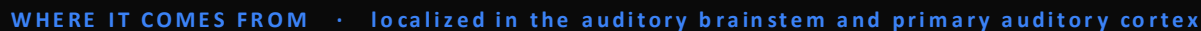
Both land in one critical band

Interference kernel, peaking at 0.25 ERB



Troughs line up with simple ratios: there the interval is wide enough that non-coincident harmonics fall outside each other's critical bands. *Plomp & Levelt, 1965 · Sethares, 1993*

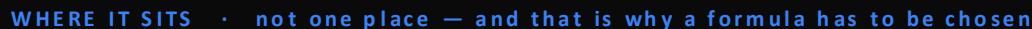
THE LANDSCAPE AXIS · H_{01} , two-pass sibling projection, $N = 16$, $\rho = 0.4$



Pooled intervals peak at T → virtual root



THE LANDSCAPE AXIS · $C_{\text{field}} = H_{01} - 1.35 R_{01} + H_{01} \cdot R_{01}$, attractors near 2:1, 3:2, 4:3, 5:4



We built this landscape into an instrument

Conchordal, running scenario autumn cycle

Conchordal is a software instrument,
running in real time.

Its sounds are not placed notes. They are agents that live, move and die in this landscape.

The bottom panels are the quantities of the last three slides — the consonance landscape, roughness and harmonicity — computed live from the sound.

The assays that follow are its minimal core, cut out for measurement.

Three minimal mechanisms

Deliberately minimal, so the landscape's contribution is attributable.

1 Pitch adaptation

Local proposals, scored leave-one-out.

$$S = C_{\text{field}} - \lambda_C \cdot \Sigma K_{\text{crowd}} - \lambda_M |\Delta x|$$

Metropolis acceptance under a decaying temperature. The crowding kernel is the analytic complement of the roughness kernel.

2 Metabolic policy

Consonance acts as an ecological resource.

$$E \leftarrow E - c_b \Delta t + r_E \cdot \max(0, C) \cdot \Delta t$$

Burn at a fixed rate, recharge with consonance, die at zero.

3 Temporal oscillation

One phase oscillator per agent.

$$\phi_i = \omega_i + K \sin(\theta^*_i - \phi_i)$$

Coupled to a shared field, not to each other. The interaction is bidirectional.

Each is a simplification of a full instrument subsystem. What is under test is the landscape, not the search strategy.

Four assays, four mechanism subsets

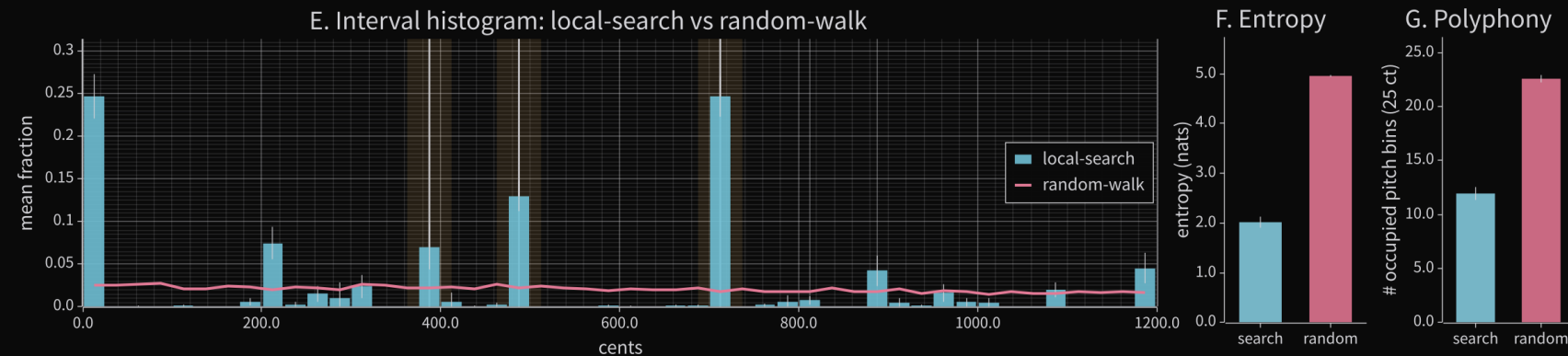
✓ = on, ✗ = off, abl. = ablated (compared with both states). All assays use a fixed 220 Hz reference.

Condition	Search	Selection	Heredity	Entrainment
Agents N	24	32	16	32
Seeds	20	20	20	20
Hill climb	abl.	✗	✗	✗
Crowding	✓	✗	✓	✗
Metabolism	✗	abl.	abl.	✗
Entrainment	✗	✗	✗	✓
Heredity	✗	✗	abl.	✗
Duration	8 sweeps	200 deaths	12k steps	40 s

Each assay isolates a different mechanism subset.

Hill-climbing yields structured polyphony

24 adaptive voices + a fixed 220 Hz drone on a 3 ct grid; hill-climbing vs. a matched random local walk.



INTERVAL ENTROPY (NATS)

2.01 vs **4.96**

Uniform ceiling 5.48.

UNIQUE PITCH BINS

11.95 vs **22.60**

24 agents settle into ≈ 12 distinct pitch classes.

JI PROXIMITY SCORE

0.366 vs **0.115**

Weighted proximity to simple ratios. Confirms the entropy result.

Each pair reads local-search first, then the matched random-walk ablation.

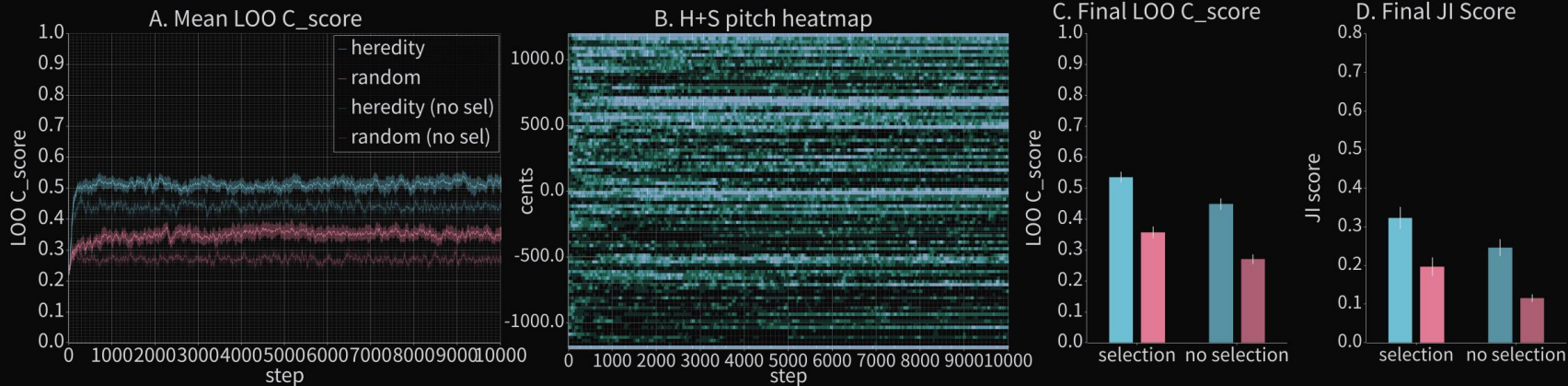


 **Listen**

Country	Percentage
Algeria	10
Argentina	10
Australia	10
Austria	10
Brazil	10
Canada	10
China	10
France	10
Germany	10
Greece	10
India	10
Indonesia	10
Italy	10
Japan	10
South Korea	10
Spain	10
Sweden	10
Switzerland	10
Taiwan	10
Thailand	10
United Kingdom	10
United States	10
South Africa	10
Other	10

Lineage bias accumulates across generations

2 × 2 factorial: respawn (heredity vs. matched log-random) × metabolic selection, adult pitch locked.



FINAL LOO C_SCORE

0.536 vs 0.358

heredity+selection vs. matched
random+selection. Heredity alone still wins
(0.449 vs. 0.271).

JI PROXIMITY / ENTROPY

0.323 / 3.28

vs. 0.197 / 4.22 for matched random.
Occupied bins contract 14.35 $\rightarrow$ 12.30:
concentration, not scatter.

SEEDS CROSSING $J_I \geq 0.50$

15 / 20

No matched-random run crosses it. What works: family heredity + local pitch re-optimisation + selection.

Hereditary adaptation you can hear

Adult pitch is locked in both arms — the only difference is how newborns are placed.



Listen

heredity + selection

Parent-biased family choice, local pitch re-optimisation, metabolic selection on.



Click to play · 12 s

matched random + selection

Log-random candidate sampling under the same scene filter; selection still on.

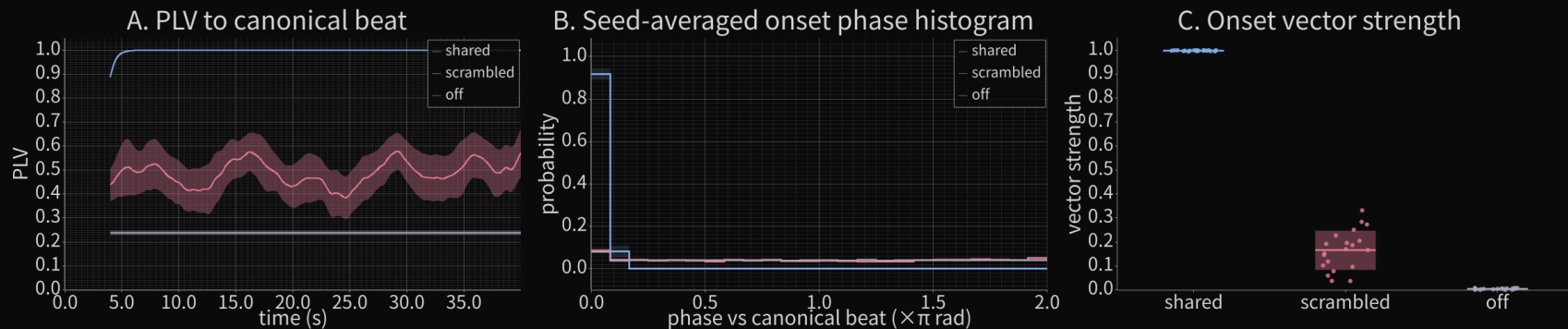


Click to play · 12 s

Only heredity+selection produces clearly musical multi-band polyphony; the controls remain diffuse. Informal listening, reported as such.

A shared field organizes timing

All agents reduced to phase oscillators, driven by an external 2 Hz scaffold.



SHARED DRIVE

0.998

All agents couple to one continuous 2 Hz phase reference.

SCRAMBLED DRIVE

0.166

Same cycle length and coupling, but phase-randomized each cycle.

DRIVE OFF

0.005

Coupling disabled; intrinsic frequencies retained.

Expected under coupled-oscillator theory. Temporal-axis validation, not an emergence claim.



The landscape, or any scalar reward?

A random permutation of consonance bins, held fixed for the run.

Same distribution of scores. Only the spatial arrangement is destroyed.

Metric	Intact landscape		Shuffled	Effect (all $p < 0.001$)
Interval entropy (nats)	2.013	→	4.416	<i>structure destroyed</i>
Unique pitch bins	11.9	→	17.0	<i>crowding over-disperses</i>
Nearest-neighbour spacing (ct)	0.576	→	0.928	<i>no collapse, but no order</i>

It is not the presence of a fitness signal that matters — a coherent psychoacoustic gradient is required.

Caveat: the hereditary assay has not been through this battery.

One psychoacoustic landscape

C: Consonance field $C_field(f)$

The plot shows the Consonance field $C_field(f)$ as a function of $\log_2(f/f_anchor)$. The x-axis ranges from -1.5 to 1.5, and the y-axis ranges from -1.0 to 1.0. The plot displays a complex landscape with several peaks and valleys, representing the psychoacoustic landscape of consonance.

Survival, movement, turnover and synchronization are all landscape position.

The same field is the system's measure of musical coherence.

Government	Percentage
Current Government	78%
Previous Government	22%

What this does not yet show

The claims the present work does not reach.

No behavioural validation

The duality predicts that ecological success tracks perceptual coherence. No listener study has been run.

A Western-leaning profile

Weighting varies across listeners and cultures. These coefficients encode one profile.

McDermott et al., 2016 — Tsimane' listeners rated consonant and dissonant chords alike.

gamelan · maqam · shruti · gagaku shō · inharmonic timbre

A fixed reference

All assays fix a 220 Hz reference — a compositional choice, not a system constraint. Remove it and the emergent tuning itself can drift.

Not open-ended

The landscape moves with the agents, but the generating rules never change. The space of what can appear is closed. Evolvable timbres and coupling topologies are the routes onward.

CONCLUSION

A viable substrate for artificial life

Self-organisation

Selection

Heredity

Entrainment

A continuous landscape derived from harmonicity and roughness sustains the core ingredients of an artificial ecology **without symbolic encoding**.

Taken together: DCC-derived consonance is a viable ALife substrate, complementing the spatial and chemical media the field has traditionally studied.

DCC is not inherently auditory

What the principle asks for is a quantitative perceptual model, not a particular sense.

The same two requirements: (i) Perceptual axes · (ii) Direct read-out

Visual

Spatial frequency and colour-opponent structure are already modelled quantitatively. A terrain built from those would stand in for art-historical rules.

Tactile

Vibrotactile perception has its own roughness. The closest formal analogue to the axis here.

Kinesthetic

Effort, balance and timing admit quantitative description. A landscape of motor-perceptual cost.

Any creative medium whose perceptual basis admits quantitative modelling could host an analogous ecology.

TRY IT

Conchordal is a playable instrument

The assays here use minimal dynamics. Each is a simplification of a subsystem in the full instrument.

For composers, it offers a generative substrate in which ecological dynamics produce musically coherent material — without symbolic specification.

Acknowledgment

The author thanks Gilberto Bernardes for helpful correspondence regarding the naming of Conchordal in relation to his Conchord system.

ALPHA TESTERS WELCOME

Composers · sound artists · ALife researchers



conchordal.org
the instrument



github.com/ktakahashi74
paper, code, audio

info@conchordal.org